



System Documentation

Electronics

Engine Control System ECS with Engine Control Unit ECU 9

- **Engine Monitoring Unit EMU 9**
- **Injector Accumulator Pressure Monitoring IAP**
- **Exhaust Aftertreatment Systems CubeBox and FlatBox**

Application: Marine

Series 4000Mx5

E532867/01E



A Rolls-Royce
solution

Engine model	Power	Application group	EGAT
12V4000M25F	115 kW/cyl.	3A, continuous operation, unrestricted, 50 Hz/1500 rpm	FlatBox
16V4000M25F	115 kW/cyl.	3A, continuous operation, unrestricted, 50 Hz/1500 rpm	FlatBox
20V4000M25F	115 kW/cyl.	3A, continuous operation, unrestricted, 50 Hz/1500 rpm	CubeBox
12V4000M25S	140 kW/cyl.	3A, continuous operation, unrestricted, 60 Hz/1800 rpm	FlatBox
16V4000M25S	140 kW/cyl.	3A, continuous operation, unrestricted, 60 Hz/1800 rpm	CubeBox
20V4000M25S	140 kW/cyl.	3A, continuous operation, unrestricted, 60 Hz/1800 rpm	CubeBox
12V4000M35F	130 kW/cyl.	3B, continuous operation, variable load, 50 Hz/1500 rpm, ICXN	FlatBox
16V4000M35F	130 kW/cyl.	3B, continuous operation, variable load, 50 Hz/1500 rpm, ICXN	FlatBox
20V4000M35F	130 kW/cyl.	3B, continuous operation, variable load, 50 Hz/1500 rpm, ICXN	CubeBox
12V4000M35S	161 kW/cyl.	3B, continuous operation, variable load, 60 Hz/1800 rpm, ICXN	FlatBox
16V4000M35S	161 kW/cyl.	3B, continuous operation, variable load, 60 Hz/1800 rpm, ICXN	CubeBox
20V4000M35S	161 kW/cyl.	3B, continuous operation, variable load, 60 Hz/1800 rpm, ICXN	CubeBox
12V4000M65R	125 kW/cyl.	1A, continuous operation, unrestricted	FlatBox
12V4000M65	140 kW/cyl.	1A, continuous operation, unrestricted	FlatBox
12V4000M65L	160 kW/cyl.	1A, continuous operation, unrestricted	FlatBox
16V4000M65R	125 kW/cyl.	1A, continuous operation, unrestricted	FlatBox
16V4000M65	140 kW/cyl.	1A, continuous operation, unrestricted	CubeBox
16V4000M65L	160 kW/cyl.	1A, continuous operation, unrestricted	CubeBox
20V4000M65	140 kW/cyl.	1A, continuous operation, unrestricted	CubeBox
20V4000M65L	160 kW/cyl.	1A, continuous operation, unrestricted	CubeBox

Table 1: Applicability

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			8.9	XZ00E50001483	739

1 Safety

1.1 Important provisions for electronic products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Assembly, maintenance, and repair without ESD („Electro Static Discharge“) equipment and ESD work station
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

1.2 Intended use of electronic products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original configuration of the delivery or in a configuration approved in writing by the manufacturer
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.